Stage 6 | Understanding Questions

Rank averaging, permutation importance, and TP53

Q1. Why rank-average, and why does the ensemble win every fold?

Checking the four-patient example. In Section 9.1, elastic-net scores for patients A–D are (1.2, –0.3, 0.4, 2.0), and XGBoost scores are (0.9, 0.1, 0.8, 0.5). Assign rank 1 to the lowest risk and rank 4 to the highest, then divide each rank by 4. The ranks and averages are:

Patient	Elastic-net score / rank	XGBoost score / rank	Mean rank
A	1.2 / 0.75	0.9 / 1.00	(0.75 + 1.00)/2 = 0.875
B	–0.3 / 0.25	0.1 / 0.25	(0.25 + 0.25)/2 = 0.250
C	0.4 / 0.50	0.8 / 0.75	(0.50 + 0.75)/2 = 0.625
D	2.0 / 1.00	0.5 / 0.50	(1.00 + 0.50)/2 = 0.750

The ensemble orders patients from highest to lowest risk as A, D, C, B. If we average raw scores instead, the elastic-net values have a wider spread (approximately –1.3 to 3.0 in the final test predictions) than the XGBoost values, so its numerical scale could disproportionately influence the average. Ranking puts both models on the same scale. The IPCW C-index depends on order, not on score magnitude, so converting an individual model to ranks preserves its own concordance, although averaging the ranks can change the combined ordering.

Stability. The latest CSV reports XGBoost paired SD 0.0023 versus ensemble E1 paired SD 0.0012, both measured for each model's difference from elastic-net; E1 gains +0.0032 over elastic-net and wins 15/15 folds, while XGBoost gains +0.0026 and wins 14/15. The models' out-of-fold Spearman correlation is approximately 0.963, so they usually agree, but not perfectly: averaging can soften a model-specific ranking error when the other model disagrees in a helpful direction. E1's advantage is therefore its consistent improvement over the elastic-net, not a large gain over XGBoost itself; its mean edge over XGBoost is only 0.0006 and is not a win on every fold.

Q2. Reading the permutation importance

Bone-marrow blasts have a strong **single-variable IPCW C-index of 0.6157 (about 0.616)**, the strongest raw clinical variable in Stage 2. Yet when the bone-marrow blast feature alone is permuted in held-out centres, the elastic-net loses only **0.0017** C-index points (XGBoost loses 0.0033). This is not a contradiction: the IPSS-R score already incorporates blast percentage through its blast-category points, so scrambling the separate blast column leaves much of the same prognostic information available to the model.

Permutation importance asks how much predictive performance is lost when *one feature group* is disrupted while other features remain unchanged; it does not measure that feature's total clinical importance or its causal effect. If we shuffled both raw blasts and the IPSS-R score, we would expect a larger decline because two overlapping routes for blast information would be disrupted. However, the exact combined drop cannot be obtained by adding their separate drops (0.0017 + 0.0178), since IPSS-R also contains haemoglobin, platelets, neutrophils and cytogenetics; a joint permutation experiment would be needed to quantify it.

Q3. TP53 in the two models

The permutation table shows a **TP53-group drop of 0.0155 for elastic-net** and **0.0067 for XGBoost**, or about 0.016 versus 0.007 when rounded. The Stage 3 lesion frequencies also show major centre differences: TP53 multi-hit occurs in 6.7% of training patients versus 18.9% of test patients, and complex karyotypes in 10.9% versus 31.3% among patients with usable karyotypes. These are separate prevalence statistics, not direct evidence of how frequently the two findings co-occur in individual patients.

Elastic-net Cox combines features additively in its linear risk predictor, so explicit TP53 variables provide a direct route for encoding their association with hazard. XGBoost can instead learn conditional rules such as “if complex karyotype and high variant allele fraction, assign higher risk,” potentially reconstructing part of the information correlated with TP53 even after the TP53 feature group is shuffled. The smaller XGBoost permutation drop therefore means that its TP53 feature group is **less irreplaceable given the other predictors**, not that TP53 biology is unimportant or that XGBoost ignores TP53.

This resembles the Stage 4 SF3B1 confounding discussion: a strong association observed by itself need not represent unique information after other predictors are considered. The distinction is that the SF3B1 example concerns an adjusted Cox association, whereas TP53 permutation importance concerns predictive reliance in the presence of correlated features. Neither analysis alone establishes a causal effect.